General distinctiveness as an estimate of variation

- To investigate the effects of information-theoretic constraints on speech variation in an unsupervised way:
  - How does the contextual information content (**surprisal**) of the bigram, given its previous context in the word, affect the phone2's distinctiveness to phone1?

$$\mathrm{I}(\text{bigram} \mid \text{context}) = -\log(\mathrm{P}(\text{bigram} \mid \text{context}))$$

$$\mathrm{I}(\textbf{kæ} \mid \textbf{\#\_\_}) = -\log(\mathrm{P}(\textbf{kæ} \mid \textbf{\#\_\_}))$$
$$\mathrm{I}(\textbf{æt} \mid \textbf{k\_\_}) = -\log(\mathrm{P}(\textbf{æt} \mid \textbf{k\_\_}))$$
$$\mathrm{I}(\textbf{ts} \mid \textbf{kæ\_\_}) = -\log(\mathrm{P}(\textbf{ts} \mid \textbf{kæ\_\_}))$$

bigram 2

phone1 **phone2**

cats: k æ t s

phone1 **phone2** **phone1** **phone2**

bigram 1 **bigram 3**

Perceptual recoverability, articulatory efficiency & speech flexibility

- Would information-theoretic constraints behave differently in the articulatory & acoustic domains?